

Non-Gaussian Tilts and Gaussian Rigidity in Stereographic Infinite-Volume Limits

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Status. Candidate full resolution, likely valid; human review required. The unrestricted compactness construction contains explicit non-Gaussian limits, but every translation-invariant limit under the source's uniform L^2 hypothesis is the free massive Gaussian. Novelty is provisional.

1 Source problem and precise answer

Zahariev constructs infinite-volume probability measures on $\mathcal{D}'(\mathbb{R}^d)$ by transporting normalized density perturbations of Gaussian measures on the sphere through stereographic projection and scaling. The abstract of [1] asks whether this construction contains non-Gaussian measures:

INFINITE VOLUME LIMITS IN EUCLIDEAN QUANTUM FIELD THEORY VIA STEREOGRAPHIC PROJECTION

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ABSTRACT. We present a general infinite volume limit construction of probability measures obeying the Glimm-Jaffe axioms of Euclidean quantum field theory in arbitrary space-time dimension. In particular, we obtain measures that may be interpreted as corresponding to scalar quantum fields with arbitrary bounded continuous self-interaction. It remains however an open problem whether this general construction contains non-Gaussian measures.

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Figure 1: The exact open problem in the abstract of [1].

There are two natural readings because Proposition 4.1 of the source allows arbitrary densities satisfying norm bounds, whereas its Euclidean-invariant Glimm–Jaffe output imposes symmetry and reflection-positive structure. We settle both:

1. In the literal unrestricted class, the answer is *yes*: a bounded cosine tilt gives an explicit non-Gaussian limit.

2. In the advertised translation-invariant/Glimm–Jaffe class, the answer is *no*: uniform L^2 contiguity and ergodicity force the limit to be the free massive Gaussian.

2 Setup from the source

Let $I_k = U_\alpha U_{\beta_k} : \mathcal{D}(\mathbb{R}^d) \rightarrow \mathcal{D}(\mathbb{S}^d)$ be the source’s test-function transport and let

$$R_k : \mathcal{D}'(\mathbb{S}^d) \longrightarrow \mathcal{D}'(\mathbb{R}^d), \quad (R_k \phi)(f) = \phi(I_k f),$$

be its transpose. Write γ_k^S for the centered sphere Gaussian with smoothed covariance $\tilde{C}_{S,k}$, and suppose

$$\inf_k \|\rho_k\|_{L^1(\gamma_k^S)} \geq K_1 > 0, \quad \sup_k \|\rho_k\|_{L^2(\gamma_k^S)} \leq K_2 < \infty. \quad (1)$$

The normalized interacting and transported measures are

$$\mu_k^S = N_k \rho_k \gamma_k^S, \quad N_k^{-1} = \int \rho_k d\gamma_k^S, \quad \nu_k = (R_k)_\# \mu_k^S, \quad \bar{\gamma}_k = (R_k)_\# \gamma_k^S.$$

The exact covariance identity and mollifier estimate in the free-field remark of [1] imply

$$\bar{\gamma}_k \implies \gamma, \quad \hat{\gamma}(f) = \exp \left[-\frac{1}{2} \langle f, (-\Delta + m^2)^{-1} f \rangle_{L^2} \right], \quad (2)$$

where $m > 0$. Proposition 4.1 gives subsequential limits of ν_k .

3 Proof intuition

There is an easy way to preserve a local non-Gaussian perturbation: make the sphere density depend on one field coordinate that is transported to a fixed Euclidean test function. This proves the literal positive answer, but it chooses a location and breaks Euclidean invariance.

The symmetry requirement changes everything. The source’s Cauchy–Schwarz estimate says that each interacting measure is uniformly L^2 -dominated by its Gaussian reference. Because both sequences have been transported to one space, this estimate survives weak convergence and produces an L^2 density of the limit with respect to the free Gaussian. The latter is mixing under translations. A translation-invariant probability with such a density can only have the constant density one.

4 The unrestricted positive example

Proposition 1. *The hypotheses (1) admit a sequence whose transported measures converge to a non-Gaussian probability on $\mathcal{D}'(\mathbb{R}^d)$.*

Proof. Fix $0 < \varepsilon < 1$ and a real nonzero $h \in \mathcal{D}(\mathbb{R}^d)$. Define on sphere fields

$$\rho_k(\phi) = 1 + \varepsilon \cos(\phi(I_k h)).$$

This is a continuous positive density and $1 - \varepsilon \leq \rho_k \leq 1 + \varepsilon$, so (1) holds. Since $\phi(I_k h) = (R_k \phi)(h)$, for every bounded continuous F on $\mathcal{D}'(\mathbb{R}^d)$,

$$\int F d\nu_k = \frac{\int F(\Phi) (1 + \varepsilon \cos \Phi(h)) d\bar{\gamma}_k(\Phi)}{\int (1 + \varepsilon \cos \Phi(h)) d\bar{\gamma}_k(\Phi)}.$$

Both numerator and denominator converge by (2); the denominator is at least $1 - \varepsilon$. Hence

$$\nu_k \Rightarrow \nu_h, \quad \frac{d\nu_h}{d\gamma}(\Phi) = \frac{1 + \varepsilon \cos \Phi(h)}{\int (1 + \varepsilon \cos \Psi(h)) d\gamma(\Psi)}. \quad (3)$$

The Gaussian variable $X = \Phi(h)$ has positive variance. Its law under ν_h is the normal law multiplied by $1 + \varepsilon \cos x$ and renormalized. This is not Gaussian: the Radon–Nikodym ratio between two equivalent one-dimensional Gaussian laws is an exponential quadratic, whereas the positive bounded oscillatory function $1 + \varepsilon \cos x$ is not. Thus ν_h is non-Gaussian. $\square$

Remark 2. *The density in Proposition 1 is deliberately non-invariant and does not claim to satisfy the source's special reflection-positivity hypotheses. It answers exactly the unrestricted Proposition 4.1 reading.*

5 Uniform domination survives the limit

Lemma 3. *Suppose along a subsequence that $\nu_k \Rightarrow \nu$ and $\bar{\gamma}_k \Rightarrow \gamma$. Under (1), there is $r \in L^2(\gamma)$ such that*

$$r \geq 0, \quad \int r d\gamma = 1, \quad \nu = r\gamma, \quad \|r\|_2 \leq K_2/K_1.$$

Proof. The Cauchy–Schwarz estimate used in the appendix of [1] gives, for every bounded continuous F on $\mathcal{D}'(\mathbb{R}^d)$,

$$\int |F| d\nu_k \leq \frac{K_2}{K_1} \left(\int |F|^2 d\bar{\gamma}_k \right)^{1/2}. \quad (4)$$

Indeed, apply the source estimate to $F \circ R_k$ before transport. Weak convergence passes (4) to the limit for bounded continuous F .

Let $\mathcal{C}$ be the linear span of bounded continuous cylinder functions. The functional $L(F) = \int F d\nu$ on $\mathcal{C}$ is therefore bounded in $L^2(\gamma)$ with norm at most K_2/K_1 . Cylinder functions are dense in $L^2(\gamma)$, so L extends uniquely. Riesz representation gives $r \in L^2(\gamma)$ with $L(F) = \int Fr d\gamma$. Positivity of L gives $r \geq 0$, and $L(1) = 1$ gives $\int r d\gamma = 1$. Taking the cylinder functions $F(\Phi) = e^{i\Phi(f)}$ shows that ν and $r\gamma$ have the same characteristic functional. Probability measures on $\mathcal{D}'(\mathbb{R}^d)$ are determined by that functional, hence $\nu = r\gamma$. $\square$

6 Mixing of the massive free field

Let τ_a translate distributions by $a \in \mathbb{R}^d$. The convention is irrelevant as long as $\Phi(f) \circ \tau_a = \Phi(f_a)$ for the oppositely translated test function.

Lemma 4. *The Gaussian probability γ in (2) is strongly mixing under translations. In particular, its translation-fixed subspace in $L^2(\gamma)$ consists of the constants.*

Proof. For $f, g \in \mathcal{D}(\mathbb{R}^d)$, the cross-covariance after a translation is, up to the Fourier normalization,

$$\langle f, (-\Delta + m^2)^{-1} g_a \rangle = \int_{\mathbb{R}^d} e^{ip \cdot a} \frac{\overline{\widehat{f}(p)} \widehat{g}(p)}{|p|^2 + m^2} dp.$$

The integrand without the phase is in L^1 , so this tends to zero as $|a| \rightarrow \infty$ by Riemann–Lebesgue. The Gaussian characteristic-function formula then shows that correlations of cylinder exponentials

$e^{i\Phi(f)}$ and their translates factor in the limit. Their linear span is dense in $L^2(\gamma)$; approximation proves strong mixing for all L^2 functions.

If $q \in L^2(\gamma)$ is translation fixed, mixing gives $\|q\|_2^2 = \langle q, q \circ \tau_a \rangle \rightarrow |\int q d\gamma|^2$. Thus q has zero variance and is constant. $\square$

7 Gaussian rigidity and the full answer

Theorem 5. *Let ν be any subsequential scaling limit in the construction of [1] under (1). If ν is translation invariant, then*

$$\nu = \gamma,$$

the centered massive free Gaussian measure with covariance $(-\Delta + m^2)^{-1}$.

Proof. Lemma 3 gives $\nu = r\gamma$ with $r \in L^2(\gamma)$. Both ν and γ are translation invariant. Radon–Nikodym uniqueness therefore gives $r \circ \tau_a = r$ in $L^2(\gamma)$ for every $a \in \mathbb{R}^d$. Lemma 4 makes r constant, and $\int r d\gamma = 1$ makes that constant equal to one. $\square$

The source proves that $O(d+1)$ -invariant sphere densities yield Euclidean-invariant limits. Its bounded continuous self-interactions have this symmetry. Theorem 5 therefore says that every such output under the exact uniform L^2 hypotheses is the free field. In particular, the source’s conjectured recovery of a non-Gaussian $P(\phi)_2$ plane measure cannot occur without leaving those exact hypotheses or changing the reference scheme.

8 Verification, scope, and novelty

The argument does not assume that absolute continuity is automatically closed under weak limits; Lemma 3 uses the uniform L^2 estimate that makes it closed here. Translation invariance is essential, as Proposition 1 demonstrates.

Bounded exact-title, exact-question, citation, non-Gaussianity, and stereographic-scaling searches through 13 August 2026 found no explicit later resolution of this scheme. Duch–Dybalski–Jahandideh [2] construct a non-Gaussian $P(\Phi)_2$ limit from growing spheres via stochastic quantization and different moment estimates. Dybalski [3] computes nontrivial connected functions after a different field renormalization. Neither supplies the uniform- L^2 classification above. Novelty should nevertheless be regarded as provisional pending specialist review.

9 Human-review recommendation

Review this as a full resolution. The highest-value checks are: (i) the transpose transport in Proposition 1; (ii) passage of (4) to the limit and the cylinder-density argument; and (iii) the scope boundary between Proposition 4.1 and the source’s Euclidean-invariant Glimm–Jaffe theorem.

References

- [1] S. Zahariev, *Infinite volume limits in Euclidean quantum field theory via stereographic projection*, arXiv:1701.05569 (2017; revised 2018).
- [2] P. Duch, W. Dybalski, and A. Jahandideh, *Stochastic quantization of two-dimensional $P(\Phi)$ Quantum Field Theory*, arXiv:2311.04137 (2023).
- [3] W. Dybalski, *Interacting Euclidean quantum fields from the Gaussian free field*, arXiv:2502.07546 (2025).